Marco Bellagente

Curriculum Vitae

Kaiserstr. 59, Heidelberg
marco.bellagente@gmail.com
https://marcobellagente93.github.io

Work experience

During my PhD I developed deep-learning tools for particle physics. My research activity focused in particular on:

- generative adversarial networks and normalizing flows;
- bayesian neural networks;
- topological methods for ML.

The results of my work are available either as publications here or (partially) on my git-hub profile here.

Technical contributions

- designing of projects structure and objectives;
- coordination with group members ($\sim$ 2-8 people) via meetings and reports;
- data preparation (simulation and preprocessing) for trainings;
- writing, running and maintaining code for neural network training.

Education

2019- **Ph.D.**, *University of Heidelberg and Max Planck Institute for Nuclear Physics*.

2017-2018 **M.Sc. (Physics)**, *University of Milan*, 110/110 cum Laude.

Schools EEML (on-line, 2021); A crash course in Machine Learning (Heidelberg, 2019); Nordic Probabilistic AI school (Trondheim, 2019);

Languages

Coding Python, C++, Git-hub, shell-scripting, LaTeX

ML Pytorch, Keras/TF, Jax/Flax, Haiku

Human English (fluent), German (intermediate), Italian (mother tongue)

Publications

2021 **Latent Space Refinement for Deep Generative Models**
R. Winterhalder, M. Bellagente and B. Nachman
`https://arxiv.org/pdf/2106.00792.pdf`, (submitted to NIPS)

2021 **Understanding Event-Generation Networks via Uncertainties**
M. Bellagente, M. Luchmann, M. Haussmann and T. Plehn,
`https://arxiv.org/pdf/2104.04543.pdf`, (accepted at INN+ workshop ICML 2021)

2020 **Invertible Networks or Partons to Detector and Back Again**
M. Bellagente, A. Butter, G. Kasieczka, T. Plehn, A. Rousselot, R. Winterhalder, L. Ardizzone and U. Köthe
https://www.scipost.org/SciPostPhys.9.5.074

2019 **How to GAN Away Detector Effects**
M. Bellagente, A. Butter, G. Kasieczka, T. Plehn and R. Winterhalder,
https://www.scipost.org/SciPostPhys.8.4.070

Personal interests

Hiking; jogging; gardening; reading; cooking everything I find interesting (I often try classics from the Japanese cuisine. Results are debatable); studying German, subsequently complaining about how bad my German is.